

AN INVARIANCE PRINCIPLE FOR SEQUENTIAL NONPARAMETRIC TEST STATISTICS UNDER CONTIGUOUS ALTERNATIVES

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SUMMARY

Nonparametric sequential test statistics based on serial ranks are considered. Invariance principles are derived under contiguous alternatives in the problems of symmetry, randomness and bivariate independence.

1. Introduction

Consider a situation in which observations $X_1, X_2, \dots$ are sequentially generated in accordance with some probability measure $P_\theta, \theta \in \Theta$. We are interested in sequential procedures for testing hypotheses regarding θ . To be more specific suppose that P_θ is determined by a density function f which is symmetric around zero,

$$P_\theta(X_1 \leq x) = \int_{-\infty}^x f(y - \theta) dy, \quad -\infty < x, \theta < \infty,$$

and that $X_1, X_2, \dots$ are statistically independent. The hypothesis of interest is $H_0: \theta = 0$. If f is known the locally most powerful test of H_0 against $H_1: \theta > 0$ is based on a statistic of the form

$$(1.1) \quad \sum_{i=1}^N \text{sign}(X_i) \frac{f'(|X_i|)}{f(|X_i|)}.$$

In this paper we are concerned with the case where f is not fully specified. Here the usual procedure is to consider tests of H_0 based on linear rank statistics such as

$$(1.2) \quad L_N = \sum_{i=1}^N \text{sign}(X_i) a_N(R_{Ni}^+)$$

where R_{Ni}^+ denotes the rank of $|X_i|$ among $|X_1|, \dots, |X_N|$ and $a_N(\cdot)$ is a suitable score function.

Regarding their use in sequential tests of H_0 , however, the statistics L_N have some disadvantages. Firstly L_N is not of the form $\sum_{i=1}^N Y_i$ with $Y_1, Y_2, \dots$ independently distributed. Secondly L_N is somewhat cumbersome from a computational point of view since, whenever a new observation X_N arises, all the previous observations $X_1, \dots, X_{N-1}$ have to be re-ranked and new scores calculated for all of them in order to obtain L_N .

Consider, instead, the rank statistics

$$(1.3) \quad T_N = \sum_{i=1}^N \text{sign}(X_i) a_i(R_{ii}^+).$$

These are better suited to sequential computation in view of the fact that

$$T_N = T_{N-1} + \text{sign}(X_N) a_N(R_{NN}^+).$$

Furthermore T_N consists of independent summands when H_0 is true.

Some asymptotic distribution theory for the sequence $(T_N, N \geq 1)$ is considered in section 2 while section 3 sketches some applications of the results. In particular (see section 3.2) a useful result pertaining to the statistics L_N of (1.2) is obtained as a by-product of our main theorem under less stringent regularity conditions than before.

In section 4 we consider the hypothesis of randomness, i.e. we let

$$P_{\theta}(X_i \leq x) = \int_{-\infty}^x f(x - \theta c_i) dx, \quad -\infty < \theta < \infty,$$

where $c_1, c_2, \dots$ are regression constants and f is no longer assumed to be symmetric. In this case the usual rank statistics for testing $H_0: \theta = 0$ against $H_1: \theta > 0$ are of the form

$$(1.4) \quad \sum_{i=1}^N (c_i - \bar{c}_n) a_i(R_{Ni})$$

where now R_{Ni} denotes the rank of X_i among $X_1, \dots, X_N$ and $\bar{c}_N = \frac{1}{N} \sum_{i=1}^N c_i$. As an alternative to these we consider the statistics

$$(1.5) \quad \sum_{i=1}^N (c_i - \bar{c}_{i-1}) a_i(R_{ii})$$

and derive asymptotic distribution theory under suitable contiguous alternatives. Section 5 sketches some applications. In particular a genuinely sequential two-sample nonparametric test procedure is presented. We also show that the well-known Theil statistic is in fact equivalent to a linear serial rank statistic.

Finally, section 6 considers test statistics for the hypothesis of bivariate independence against a class of alternatives of the form suggested by Behnen (1971).

In virtually all previous literature on the subject, the quantities R_{ii}^+ and R_{ii}^- have been referred to as sequential ranks. In this paper we have chosen to call them *serial ranks* in order to avoid confusion with the use of the word "sequential" in the testing context. Thus, we refer to the statistics (1.3) as *linear signed serial rank statistics* and to the statistics (1.5) as *linear serial rank statistics*.

The basic method used in this paper consists in projecting a parametric test statistic (such as (1.1)) into a suitable space of rank statistics. As such, repeated use is made of the well-known *Hajek projection lemma*. In order to state it generally and in a form best suited to our use, let H denote the (Hilbert) space of real square-integrable random variables on a probability space (Ω, F, P) and let M be any closed linear subspace of H . Given $Y \in H$, the projection $\Pi(Y|M)$ of Y into M is defined as the almost sure unique member of M which satisfies the equation

$$\inf \{ \xi(Y - X)^2 : X \in M \} = \xi(Y - \Pi(Y|M))^2$$

with ξ denoting expectation under P . In this case it is also true that

$$(1.6) \quad \xi(Y - \Pi(Y|M))^2 = \xi Y^2 - \xi(\Pi(Y|M))^2.$$

(In particular, if B is a sub-sigma field of F and if B^* is the class of B -measurable members of H , then $\Pi(Y|B^*) = \xi(Y|B)$, the conditional expectation of Y given B .)

Now let $N \geq 1$ be given and let $Z_1, \dots, Z_N$ be independent k -dimensional random vectors on (Ω, F, P) . Let M consist of all real random variables of the form $\sum_{i=1}^N g_i(Z_i)$

where each g_i is real-valued and measurable and $\xi(g_i(Z_i))^2 < \infty$ for $i = 1, \dots, N$. The content of the Hajek Projection Lemma is that

$$(1.7) \quad \Pi(Y|M) = \sum_{i=1}^N \xi(Y|Z_i) - (N-1) \xi(Y).$$

Reynolds (1975) has considered a sequential test of symmetry around zero and derived its asymptotic power and expected sample size under a *fixed* alternative. His test is based on the statistics $\sum_{i=1}^N \text{sign}(X_i) \frac{R_{ii}^+}{i+1}$ which are obtained from (1.3) by letting $a_i(j) =$

$\frac{1}{i+1}$ for $i \leq j \leq i$. Our results complement those of Reynolds to the extent that we consider a wider class of statistics under *contiguous* alternatives. Our method, however, is quite different from that used by Reynolds.

Assuming that $X_1, X_2, \dots$ are independent and identically distributed, Sen (1978b) derives almost sure invariance principles for statistics of the form $\sum_{i=2}^N g(X_i) a_i(R_{ii})$, which he terms "rank-discounted partial sums". In view of the assumptions he makes regarding g , these statistics do not seem well suited for use in testing the randomness hypothesis.

More recently, Mason (1981) has used properties of the statistics in (1.3) and (1.5) to derive results on the speed of convergence to normality, Berry-Esseen-type theorems and laws of the iterated logarithm for the usual linear rank statistics *under the respective null hypotheses*. A further contribution in this direction is contained in sections 3.2 and 5.2 of the present paper.

We assume familiarity with the concept of weak convergence of probability distributions on the space $D[0, \infty)$, a good reference being Lindvall (1973). Contrary to normal usage, however, we conceive of a $D[0, \infty)$ populated by *left-continuous* functions. For convenient reference, a standard result of the theory is stated in the appendix (section 7). the symbol $\Rightarrow$ denotes weak convergence and is always subscripted by the appropriate probability symbol. For example, $\Rightarrow_{Q_N}$ would denote weak convergence under a sequence of probability measures $\{Q_N\}$. Throughout the paper, $(W(t), t \geq 0)$ will denote a standard Wiener process. We also assume familiarity with the theory of contiguous sequences of probability measures as it appears in Hajek & Sidak (1967). In particular, frequent use is made of the result that if $\{P_N\}$ is a tight sequence and $\{Q_N\}$ is contiguous to $\{P_N\}$, then $\{Q_N\}$ is also tight.

2. The symmetry hypothesis

Let f be an absolutely continuous density function which is symmetric around 0 and has finite Fisher information

$$(2.1) \quad I(f) = \int_{-\infty}^{+\infty} (f'(x) / f(x))^2 f(x) dx < \infty.$$

Let P be a probability measure under which the random variables $X_1, X_2, \dots$ are independent and identically distributed with common density function f . Denote by $|X|_N^{(i)}$

the i 'th order statistic of $|X_1|, \dots, |X_N|$, let $R_{Ni}^+ = \sum_{j=1}^N I(|X_j| \leq |X_i|)$ where $I(\cdot)$ is the indicator function and let $\text{sign}(X_i) = 2I(X_i > 0) - 1$, $1 \leq i \leq N$.

Lemma 2.1

Under P it holds that

$$(i) \quad \text{sign}(X_i) \text{ and } |X_i| \text{ are independent; } i \geq 1$$

(ii) $\text{sign}(X_i), |X_N^{(j)}|$ and R_{Nk}^+ are independent; $1 \leq i, j, k \leq N, N \geq 1$

(iii)* $\text{sign}(X_i), |X_N^{(j)}|$ and R_{kk}^+ are independent; $1 \leq i, j, k \leq N, N \geq 1$

(iv) $R_{11}^+, R_{22}^+, \dots$ are independent and

$$P(R_{NN}^+ = j) = N^{-1} \text{ for } j = 1, \dots, N; N \geq 1.$$

Proof

See Reynolds (1975). ■

Now let $\{a_i(j), 1 \leq j \leq i; i \geq 1\}$ be a triangular array of constants such that

$$(2.2) \quad \lim_{i \rightarrow \infty} \int_0^1 (a_i(1 + [ui]) - \phi(u))^2 du = 0$$

where $[y]$ denotes the integer part of y and ϕ is a real function on $(0,1)$ satisfying the condition

$$(2.3) \quad A^2 = \int_0^1 \phi^2(u) du < \infty.$$

Consider the linear signed serial rank statistics

$$(2.4) \quad T_N = \sum_{i=1}^N c_i \text{sign}(X_i) a_i(R_{ii}^+)$$

where $c_1, c_2, \dots$ are constants satisfying for $N \rightarrow \infty$ the condition

$$(2.5) \quad \max \{c_i^2 : 1 \leq i \leq k_N(t)\} = o(C(N))$$

with

$$(2.6) \quad \begin{aligned} k_N(t) &= \min \{k \geq 1 : C^2(k) \geq tC^2(N)\} \\ C^2(N) &= \sum_{i=1}^N c_i^2. \end{aligned}$$

Note that (2.5) implies for $N \rightarrow \infty$ that

$$(2.7) \quad C^2(N) \rightarrow \infty$$

and

$$(2.8) \quad C^2(k_N(t) / C^2(N)) \rightarrow t, \text{ for every } t > 0.$$

Using (2.2), (2.7) and Lemma 2.1 we easily find that, under P ,

$$(2.9) \quad \begin{aligned} \xi(T_N) &= 0 \\ \text{var}(T_N) / A^2 C^2(N) &\rightarrow 1. \end{aligned}$$

Next, let F_+ denote the distribution function of $|X_1|$ and write $U_1 = F_+(|X_1|)$.

Then $U_1, U_2, \dots$ is a sequence of independent and uniformly distributed (over $(0,1)$) random variables and we denote by $U_i^{(j)}$ the j 'th order statistic of $U_1, \dots, U_i$. Also let

$$(2.10) \quad \begin{aligned} T_N^\phi &= \sum_{i=1}^N c_i \text{sign}(X_i) \phi(U_i) \\ \hat{T}_N^\phi &= \sum_{i=1}^N c_i \text{sign}(X_i) a_i^\phi(R_{ii}^+) \end{aligned}$$

where, as usual, $a_i^\phi(j) = \xi\phi(U_i^{(j)})$.

Lemma 2.2

Let L_N^+ denote the class of linear signed serial rank statistics, that is, statistics of the form $\sum_{i=1}^N g_i(\alpha_i, \text{sign}(X_i), R_{ii}^+)$ where $\alpha_1, \alpha_2, \dots$ are finite constants and $\xi g_i^2(\cdot) < \infty$.

Let (2.1) and (2.3) hold. Then

$$\hat{T}_N^\phi = \Pi(T_N^\phi | L_N^+).$$

Proof

Put $Z_i = (\alpha_i, \text{sign}(X_i), R_{ii}^+)$ in (1.7) and apply Lemma 2.1.

■

The above Lemma, in conjunction with (1.6) and the relation

$$(2.11) \quad \lim_{i \rightarrow \infty} \int_0^1 (a_i^\phi(1 + [ui]) - \phi(u))^2 du = 0$$

(see Hajek & Sidak (1967, page 158)), results in

$$\begin{aligned}
 (2.12) \quad & \xi(T_N^\phi - \hat{T}_N^\phi)^2 \\
 &= C^2(N)A^2 - \sum_{i=1}^N c_i^2 \int_0^1 (a_i^\phi(1 + [ui]))^2 du \\
 &= o(C^2(N)).
 \end{aligned}$$

Also, one may prove directly from (2.2), (2.7) and (2.9) that

$$(2.13) \quad \xi(T_N - \hat{T}_N^\phi)^2 = o(C^2(N)).$$

Lemma 2.3

If (2.1) through (2.5) hold, then for every $t > 0$, $\max \{ |T_k - T_k^\phi| : 1 \leq k \leq k_N(t) \} = o(C(N))$ in probability under P .

Proof

Let F_N be the σ -field generated by $X_1, \dots, X_N$. Then $\{(T_N - T_N^\phi; F_N), N \geq 1\}$ is a martingale. Application of the maximal inequality together with (2.12), (2.13) and (2.7) then yields the required result. ■

Let $Q_{N,\beta}$, for every β and every $N \geq 1$, be a probability measure under which X_1 has the density function

$$(2.14) \quad q_{N,\beta}^{(i)}(x) = f(x - d_i\beta) / \left(\sum_{i=1}^N d_i^2 \right)^{1/2}$$

and $X_1, X_2, \dots$ are independently distributed. Here $d_1, d_2, \dots$ are constants which for every $t > 0$ satisfy the following conditions as $N \rightarrow \infty$:

$$(2.15) \quad \max \{ d_i^2 : 1 \leq i \leq k_N(t) \} = o\left(\sum_{i=1}^N d_i^2 \right)$$

$$(2.16) \quad \sum_{i=1}^{k_N(t)} c_i d_i / \left(\sum_{i=1}^N c_i^2 \sum_{i=1}^N d_i^2 \right)^{1/2} \rightarrow s(t)$$

and

$$(2.17) \quad (s(t), t \geq 0) \in D[0, \infty).$$

Also let

$$(2.18) \quad e(\phi, f) = \int_0^1 \phi(u) \phi_+(u, f) du / A$$

where

$$\phi_+(u, f) = -f'(F_+^{-1}(u)) / f(F_+^{-1}(u)), 0 < u < 1,$$

Theorem 1

Let (2.1), (2.2), (2.3), (2.5) and (2.15), (2.16), (2.17) hold. Then, for every β ,

$$(2.19) \quad (T_{k_N(t)} / AC(N), t \geq 0) \Rightarrow_{Q_{N,\beta}} (W(t) + \beta e(\phi, f) \rho(ts(t))^{1/2}, t \geq 0)$$

where $Q_{N,\beta}$ is defined through (2.14) and $e(\phi, f)$ by (2.18).

Proof

Under (2.15) it follows by the method in section VI 2.1 of Hajek & Sidak (1967) that the $Q_{N,\beta}$ -distributions of the vectors $(X_1, \dots, X_{k_N(\tau)})$ are for every $\beta > 0$, and $\tau > 0$ contiguous to their $Q_{N,0}(=P)$ -distributions. This together with lemma 2.3, implies that

$$\sup \{ |T_{k_N(\tau)} - T_{k_N(\tau)}^\phi| : 0 \leq \tau \leq t \} = o(C(N))$$

in probability under $Q_{N,\beta}$ for every $\beta \neq 0$ and $t > 0$. Thus, in order to prove the theorem, it is sufficient to show that

$$(T_{k_N(\tau)}^\phi / AC(N), 0 \leq \tau \leq t) \Rightarrow_{Q_{N,\beta}} (W(\tau) + \beta e(\phi, f) s(\tau), 0 \leq \tau \leq t)$$

for every β and every $t > 0$.

For $\beta = 0$ this follows immediately from theorem 3 in the appendix. The consequent tightness of the sequence of $Q_{N,\beta}$ -distributions of the processes

$$(T_{k_N(\tau)}^\phi / AC(N), 0 \leq \tau \leq t) \text{ for } \beta = 0,$$

together with contiguity, implies its tightness for all $\beta \neq 0$. It remains therefore to show that the finite dimensional distributions converge correctly, i.e.

$$Z_N \equiv \sum_{k=1}^m \lambda_k T_{k_N(\tau_k)}^\phi / AC(N) \Rightarrow_{Q_{N,\beta}} \sum_{k=1}^m \lambda_k (W(\tau_k) + \beta e(\phi, f) \rho(\tau_k s(\tau_k))^{1/2})$$

for all $m \geq 1$, $-\infty < \lambda_1, \dots, \lambda_m < \infty$ and $0 \leq \tau_1 < \dots < \tau_m \leq t$. The easiest way of accomplishing this is to use Le Cam's third lemma. Thus, we let

$$V_N = - \sum_{i=1}^{k_N(\tau_m)} \frac{\beta d_i}{\left(\sum_{i=1}^N d_i^2 \right)^{1/2}} \text{sign}(X_i) \frac{f'(X_i)}{f(X_i)}$$

and find the asymptotic joint distribution of (Z_N, V_N) under P . The details of this computation (similar to that appearing on pages 217 and 218 of Hajek & Sidak (1967)) are straightforward, though tedious. We omit them.

3. Applications

3.1 Martingale property of linear signed rank statistics

Let G_N denote the sigma field generated by $\text{sign}(X_1), \dots, \text{sign}(X_N), R_{N1}^+, \dots, R_{NN}^+$. Note that G_N is also the sigma field generated by $\text{sign}(X_1), \dots, \text{sign}(X_N), R_{11}^+, \dots, R_{NN}^+$. Then $\{(T_N, G_N), N \geq 1\}$ is clearly a P -martingale.

Next, let

$$L_N^\phi = \sum_{i=1}^N c_i \text{sign}(X_i) a_i^\phi(R_{Ni}^+).$$

It is known that $L_N^\phi = \xi(T_N^\phi | G_N)$ under P , with T_N^ϕ defined in (2.10). Hence

$$\begin{aligned} & \xi(L_N^\phi | G_{N-1}) \\ &= \xi(\xi(T_N^\phi | G_N) | G_{N-1}) \\ &= \xi(T_N^\phi | G_{N-1}) \\ &= \xi(T_{N-1}^\phi | G_{N-1}) + \xi(\text{sign}(X_N) \phi(U_N) | G_{N-1}) \\ &= L_{N-1}^\phi \end{aligned}$$

and $\{(L_N^\phi, G_N); N \geq 1\}$ is therefore a P -martingale.

This last result is not new, having been obtained previously by Sen & Ghosh (1973). The proof we have given above is somewhat different from theirs, though.

3.2 Asymptotic distribution of linear signed rank statistics

Let G_N^* denote the class of all G_N -measurable random variables. Since $G_N^* \supset L_N^+$, we have

$$\Pi(L_N^\phi | L_N^+) = \Pi(\xi(T_N^\phi | G_N) | L_N^+) = \Pi(\Pi(T_N^\phi | G_N^*) | L_N^+) = \Pi(T_N^\phi | L_N^+) = \hat{T}_N^\phi.$$

It follows that $\hat{T}_N^\phi$ is also the projection of L_N^ϕ into the class of linear signed serial rank statistics, whence

$$\begin{aligned} \xi(L_N^\phi - \hat{T}_N^\phi)^2 &= \xi(L_N^\phi)^2 - \xi(\hat{T}_N^\phi)^2 \\ &= C^2(N) \frac{1}{N} \sum_{i=1}^N (a_N^\phi(i))^2 - \sum_{i=1}^N c_i^2 \frac{1}{i} \sum_{j=1}^i (a_i^\phi(j))^2 \\ &= \alpha(C^2(N)) \end{aligned}$$

since $i^{-1} \sum_{j=1}^i (a_i^\phi(j))^2 \rightarrow A^2$ as $i \rightarrow \infty$, see Hajek & Sidak (1967, Theorem b, page 158). The maximal inequality for martingales, applied to $L_N^\phi - \hat{T}_N^\phi$, therefore yields for every $t > 0$,

$$(3.1) \quad \max \{ |L_k^\phi - \hat{T}_k^\phi| : 1 \leq k \leq k_N(t) \} = \alpha(C(N))$$

in probability under P . (We assume that the conditions of Theorem 1 are in force.) Taking account of contiguity we find that (3.1) also holds in probability under all $Q_{N,\beta}$. In particular, the asymptotic $Q_{N,\beta}$ -distributions of $\{L_{k_N(\tau)}^\phi / AC(N), 0 \leq \tau \leq t\}$ and $\{\hat{T}_{k_N(\tau)}^\phi / AC(N), 0 \leq \tau \leq t\}$ are the same for all $t > 0$. From Theorem 1 we therefore obtain

$$(L_{k_N(t)}^\phi / AC(N), t \geq 0) \xrightarrow{Q_{N,\beta}} (W(t) + \beta e(\phi, f) s(t), t \geq 0).$$

A similar result was previously obtained by Sen (1977, Theorem 3.1) under more restrictive regularity conditions — see his equation (2.3).

3.3 Truncated sequential tests

Let P_θ be a probability measure under which the random variables $X_1, X_2, \dots$ are i.i.d. with common density function $f(x - \theta)$. In order to test $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$, a stopping rule such as

$$\tau_N = \min \{k \geq 1 : |T_k| \geq c_N(\alpha)\}$$

may be used. Here T_k is given by (2.4) with $c_1 = c_2 = \dots = 1$ and $c_N(\alpha)$ is a constant such that $P_0\{\tau_N \leq N\} = \alpha$, $0 < \alpha < 1$. The rule is to reject H_0 if $\tau_N \leq N$, and to accept it otherwise. Such a test has been considered by Reynolds (1975) in the special case $a_i(j) = j/(i+1)$ where $\phi(u) = u$.

In view of the simple distributional structure of T_k under H_0 , an approximate value of $c_N(\alpha)$ may be estimated from a simulation experiment. Alternatively, if $c(\alpha)$ satisfies the equation

$$(3.2) \quad \int_{c(\alpha)}^{\infty} (2\pi)^{-1/2} \exp(-t^2/2) dt = \alpha/4,$$

then $c_N(\alpha) \approx AN^{1/2} c(\alpha)$ as $N \rightarrow \infty$. This follows from (2.19):

$$\begin{aligned} P_0\{\tau_N \leq N\} &= P_0\{|T_k|/AN^{1/2} \geq c_N(\alpha)/AN^{1/2} \text{ for some } 1 \leq k \leq N\} \\ &\rightarrow P\{|W(t)| \geq c(\alpha) \text{ for some } 0 \leq t \leq 1\} \\ &\approx \alpha, \text{ see e.g. Reynolds (1975, page 394).} \end{aligned}$$

Hence, letting $c_N(\alpha) = AN^{1/2} c(\alpha)$ we obtain a test with type I error approximately α .

Next, we define a local asymptotic power function for this sequential test procedure by

$$\gamma(\beta) = \lim_{N \rightarrow \infty} P_{\beta N^{-1/2}}(\tau_N \leq N).$$

Letting $\mu = \int_0^1 \phi(u) \cdot \phi(u, t) \cdot du / A$, we obtain from Theorem 1, with $C(N) = N^{1/2}$,

$$\begin{aligned} \gamma(\beta) &= \text{Prob}\{|W(t) + \Delta t \mu| \geq c(\alpha) \text{ for some } 0 \leq t \leq 1\} \\ &\approx \exp(2\Delta \mu c(\alpha)) \Phi(-\Delta \mu - c(\alpha)) + \Phi(\Delta \mu - c(\alpha)) \\ &\quad + \exp(-2\Delta \mu c(\alpha)) \Phi(\Delta \mu - c(\alpha)) + \Phi(-\Delta \mu - c(\alpha)). \end{aligned}$$

If $\gamma_N(\theta) = P_{\theta}\{\tau_N \leq N\}$ denotes the usual power function of the test, we therefore approximately have $\gamma_N(\theta) \approx \gamma(\theta N^{1/2})$.

4. The randomness hypothesis

In this section we drop the symmetry assumption on f , but retain (2.1). The i 'th order statistic of $X_1, \dots, X_N$ is denoted by $X_N^{(i)}$ and the rank of X_i is $R_{Ni} = \sum_{j=1}^N I(X_j \leq X_i)$. In view of Lemma 2.1, the following requires no proof.

Lemma 4.1

Under P it holds that

$$(i) \quad X_N^{(i)} \text{ and } R_{Nj} \text{ are independent; } 1 \leq i, j \leq N, N \geq 2$$

$$(ii) \quad X_N^{(i)} \text{ and } R_{jj} \text{ are independent; } 1 \leq i, j \leq N, N \geq 2$$

$$(iii) \quad R_{22}, R_{33}, \dots \text{ are independent and}$$

$$P(R_{NN} = j) = N^{-1} \text{ for } j = 1, \dots, N; N \geq 2.$$

■

Let $\{ \{a_i(j), 1 \leq j \leq i\}, i \geq 1 \}$ satisfy (2.2) with ϕ subject to (2.3) and to

$$(4.1) \quad \int_0^1 \phi(u) du = 0.$$

Also let

$$(4.2) \quad \sum_{j=1}^i a_i(j) = 0 \text{ for } i \geq 2.$$

Put

$$\bar{c}_i = \frac{1}{i} \sum_{j=1}^i c_j \text{ for } i \geq 1$$

$$(4.3) \quad b_i = c_i - \bar{c}_{i-1} \text{ for } i \geq 2$$

$$B^2(k) = \sum_{i=2}^k b_i^2 \text{ for } k \geq 2$$

and, for $N \geq 2$, let

$$(4.4) \quad m_N(t) = \min \{k \geq 2 : B^2(k) \geq tB^2(N)\}.$$

Assume that as $N \rightarrow \infty$

$$(4.5) \quad \max \{b_i^2 : 2 \leq i \leq m_N(t)\} = o(B^2(N))$$

for every $t > 0$, and note that this implies

$$(4.6) \quad B^2(N) \rightarrow \infty$$

and

$$(4.7) \quad B^2(m_N(t))/B^2(N) \rightarrow t \text{ for every } t > 0.$$

In order to study the asymptotic distribution of the sequence of general linear serial rank statistics

$$(4.8) \quad T_N = \sum_{i=2}^N (c_i - \bar{c}_{i-1}) a_i(R_{ii}) \equiv \sum_{i=2}^N b_i a_i(R_{ii})$$

we introduce the statistics

$$(4.9) \quad T_N^\phi = \sum_{i=1}^N (c_i - \bar{c}_N) \phi(U_i)$$

$$\hat{T}_N^\phi = \sum_{i=2}^N (\bar{c}_i - \bar{c}_{i-1}) a_i^\phi(R_{ii}) \equiv \sum_{i=2}^N b_i a_i^\phi(R_{ii})$$

where $U_i = F(X_i)$, F denoting the common c.d.f. of the X_i 's and where $a_i^\phi(j) = \xi_\phi(U_i^{(j)})$ with $U_i^{(1)} < \dots < U_i^{(i)}$ being the order statistics of $U_1, \dots, U_i$.

Lemma 4.2

Let L_N denote the class of linear serial rank statistics, that is, statistics of the form

$$\sum_{i=2}^N g_i(\alpha_i, R_{ii}) \text{ where } \alpha_1, \alpha_2, \dots \text{ are constants and } \xi_{g_i}^2(\cdot) < \infty. \text{ Then}$$

$$\hat{T}_N^\phi = \Pi(T_N^\phi | L_N).$$

Proof

Using lemma 4.1 we obtain for each fixed $i \geq 2$ that

$$\xi_\phi\{\phi(U_i) | R_{ii}\} = a_i^\phi(R_{ii})$$

and for $k > i$

$$\xi \{ \phi(U_k) | R_{ii} \} = \xi \phi(U_k) = 0.$$

Lastly, for $k < i$,

$$\begin{aligned} & \xi \{ \phi(U_k) | R_{ii} \} \\ &= \xi \{ \xi \{ \phi(U_k) | R_{ik}, R_{ii} \} | R_{ii} \} \\ &= \xi \{ a_i^\phi(R_{ik}) | R_{ii} \}. \end{aligned}$$

But

$$\begin{aligned} & \xi \{ a_i^\phi(R_{ik}) | R_{ii} = j \} \\ &= \frac{1}{i-1} \sum_{\substack{\ell=1 \\ (\ell \neq j)}}^i a_i^\phi(\ell) \\ &= -\frac{1}{i-1} a_i^\phi(j) \end{aligned}$$

$$\text{Since } \sum_{\ell=1}^i a_i^\phi(\ell) = i \int_0^1 \phi(u) du = 0.$$

The required result now follows upon application of the projection lemma to the vectors (α_i, R_{ii}) , $i \geq 2$. ■

Noting that $\sum_{i=1}^N (c_i - \bar{c}_N)^2 / B^2(N) \rightarrow 1$ as $N \rightarrow \infty$, we obtain by a method entirely similar to that used in deriving (2.12) and (2.13) that $\xi(T_N^\phi - \hat{T}_N^\phi)^2 = o(B^2(N))$ and $\xi(T_N - \hat{T}_N^\phi)^2 = o(B^2(N))$. Hence, as $N \rightarrow \infty$,

$$(4.10) \quad T_N - \hat{T}_N^\phi = o(B(N)) \quad \text{in probability under } P.$$

Lemma 4.3

Let T_N be given by (4.8) and assume that (2.1), (2.2), (2.3), (4.1) and (4.5) hold. Then

$$(T_{m_N(t)} / AB(N), t \geq 0) \xrightarrow{P} (W(t), t \geq 0).$$

Proof

Applying the special central limit theorem of Hajek & Sidak (1967, page 153) to the sequence $\{T_N^\phi\}$ and taking account of (4.7) and (4.10), we obtain for each fixed $\tau > 0$ that

$$T_{m_N(\tau)} / AB(N) \xrightarrow{P} W(\tau).$$

Letting $Y_i = b_{i1} a_{i1}(R_{ii})$ for $i \geq 2$ and noting that $\sum_{i=1}^N \xi Y_i^2 / A^2 B^2(N) \rightarrow 1$, we have, equivalently, for each fixed $\tau > 0$ that

$$(4.11) \quad \sum_{i=2}^{m_N(\tau)} Y_i / \left\{ \sum_{i=2}^N \xi Y_i^2 \right\}^{1/2} \xrightarrow{P} W(\tau).$$

Furthermore

$$\begin{aligned} (4.12) \quad & \max \left\{ \text{var } Y_i / \sum_{i=1}^N \xi Y_i^2 : 2 \leq i \leq m_N(\tau) \right\} \\ & \sim \max \left\{ b_i^2 / B^2(N) : 2 \leq i \leq m_N(\tau) \right\} \\ & = o(1) \text{ as } N \rightarrow \infty. \end{aligned}$$

But (4.11) and (4.12) together imply, by the Normal convergence criterion (Loève 1963, page 295), that for each $\tau > 0$ the following Lindeberg condition holds, viz.

$$\sum_{i=1}^N \left(\xi Y_i^2 \right)^{-1} \sum_{i=1}^{m_N(\tau)} \xi Y_i^2 I \left(Y_i^2 \geq \varepsilon \sum_{i=1}^N \xi Y_i^2 \right) = o(1) \text{ as } N \rightarrow \infty$$

for each $\varepsilon > 0$.

Thus we obtain lemma 4.3 upon applying Theorem 4 in the Appendix to $\{Y_i, i \geq 2\}$ and noting that as $N \rightarrow \infty$,

$$\max \left\{ \left| \sum_{i=1}^k \xi Y_i^2 / \sum_{i=1}^N \xi Y_i^2 - B^2(k)/B^2(N) \right| : 2 \leq k \leq m_N(\tau) \right\} = o(1)$$

for each $\tau > 0$. ■

Let $Q_{N,\beta}$ for every $\beta \geq 0$ and $N \geq 2$ be a probability measure under which X_1 has the density function

$$(4.13) \quad q_{N,\beta}^{(i)} = f \left(x - d_1 \beta / \left(\sum_{i=2}^N d_i^2 \right)^{1/2} \right)$$

and $X_1, X_2, \dots$ are independent. Here $d_1, d_2, \dots$ are constants which for every $\tau > 0$ satisfy the following condition as $N \rightarrow \infty$, viz.

$$(4.14) \quad \max\{d_i^2 : 1 \leq i \leq m_N(\tau)\} = o\left(\sum_{i=1}^N d_i^2\right)$$

$$(4.15) \quad \sum_{i=1}^{m_N(\tau)} (c_i - c_{m_N(\tau)}) d_i / B(N) \left(\sum_{i=1}^N d_i^2\right)^{1/2} \rightarrow s(\tau)$$

$$(4.16) \quad (s(\tau), \tau \geq 0) \in D[0, \infty).$$

Also let

$$(4.17) \quad e(\phi, f) = \int_0^1 \phi(u) \phi(u, f) du / A$$

where

$$\phi(u, f) = -f'(F^{-1}(u)) / f(F^{-1}(u)), 0 < u < 1.$$

Theorem 2

Let (4.14), (4.15), (4.16) and the assumptions of lemma 4.3 hold. Then, for every β ,

$$(T_{m_N(t)}' / AB(N), t \geq 0) \xrightarrow{Q_{N,\beta}} (W(t) + \beta e(\phi, f) s(t)), t \geq 0)$$

where $Q_{N,\beta}$ is defined through (4.13) and $e(\phi, f)$ by (4.17).

Proof

Under (4.14) and (4.15), by the method of Hajek & Sidak (1967 section VI 2.1), the $Q_{N,\beta}$ -distributions of the vectors $(X_1, \dots, X_{m_N(\tau)})$ are, for every $\beta \neq 0$ and $t > 0$, contiguous to their $Q_{N,0} (= P)$ -distributions as $N \rightarrow \infty$. In virtue of lemma 4.3 therefore, for every β and every $t > 0$, the sequence of $Q_{N,\beta}$ -distributions of the processes $(T_{m_N(\tau)})$,

$0 \leq \tau \leq t$ is tight. All that is left to show is that the vectors $(T_{m_N(t_1)}, \dots, T_{m_N(t_k)}) / AB(N)$ converge to the correct multivariate normal distribution for all fixed $t_1, \dots, t_k \geq 0$

and $k \geq 1$. In view of (4.10) and contiguity, however, it suffices to consider the vectors $(T_{m_N}^\phi(t_1), \dots, T_{m_N}^\phi(t_k)) / AB(N)$ instead. As in the proof of Theorem 1 application of

Le Cam's third lemma yields the required result.

5. Applications

5.1 Martingale property of linear rank statistics

Let H_N be the sigma field generated by $R_{N2}, \dots, R_{NN}$, which is the same as the σ -field generated by $R_{22}, \dots, R_{NN}$. Then $\{(T_N^\phi, H_N), N \geq 2\}$ is clearly a P-martingale.

Next, if

$$L_N^\phi = \sum_{i=1}^N (c_i - \bar{c}_N) a_N^\phi(R_{Ni})$$

then it is known that $L_N^\phi = \xi(T_N^\phi | H_N)$ under P, where T_N^ϕ is defined in (4.9). It follows that

$$\begin{aligned} & \xi(L_N^\phi | H_{N-1}) \\ &= \xi(T_N^\phi | H_{N-1}) \\ &= \xi(T_{N-1}^\phi | H_{N-1}) + (\bar{c}_{N-1} - \bar{c}_N) \xi\left\{\sum_{i=1}^{N-1} \phi(U_i) | H_{N-1}\right\} \\ &+ (c_N - \bar{c}_N) \xi(\phi(U_N) | H_{N-1}) \\ &= L_{N-1}^\phi + (\bar{c}_{N-1} - \bar{c}_N) \sum_{j=1}^{N-1} a_{N-1}^\phi(j) + (c_N - \bar{c}_N) \xi\phi(U_N) \\ &= L_{N-1}^\phi \end{aligned}$$

$$\text{since } \sum_{j=1}^{N-1} a_{N-1}^\phi(j) = (N-1) \xi\phi(U_{N-1}) = 0 = \xi\phi(U_N).$$

It follows that $\{(L_N^\phi, H_N), N \geq 2\}$ is a P-martingale, a result which was obtained by Sen & Ghosh (1972) by an essentially different method.

5.2 Asymptotic distribution of linear rank statistics

Using essentially the same technique as in section 3.2 above it may be shown that $\hat{T}_N^\phi$ is also the projection of L_N^ϕ into L_N and that

$$\xi_0(L_N^\phi - \hat{T}_N^\phi)^2 = o(B^2(N)).$$

Taking contiguity into account this leads, as in section 3.2, to

$$\max \{ |L_k^\phi - \hat{T}_k^\phi| ; 2 \leq k \leq m_N(t) \} = o(B(N))$$

in probability under $Q_{N,\beta}$. This, together with Theorem 2, yields.

$$(L_{k_N(t)}^\phi / AB(N), t \geq 0) \xrightarrow{Q_{N,\beta}} (W(t) + \beta e(\phi, f)s(t), t \geq 0).$$

A similar result has recently been obtained by Sen (1978a, Corollary 3.1.1) using different methods.

5.3 Sequential two-sample test

Suppose that each X_i is generated by one of two c.d.f.'s G or H . We wish to test the hypothesis that $G \equiv H$ against the alternative that they differ only by a shift in location.

Let, for $i \geq 1$,

$$c_i = \begin{cases} 1 & \text{if } X_i \text{ comes from } G \\ -1 & \text{if } X_i \text{ comes from } H \end{cases}$$

and assume that $\bar{c}_N \equiv \frac{1}{N} \sum_{i=1}^N c_i \rightarrow c$ as $N \rightarrow \infty$ for some c in $(-1, 1)$. A sequential test of the null hypothesis may then be based on the statistics

$$T_N = \sum_{i=2}^N (c_i - \bar{c}_{i-1}) a_i(R_{ii}), N \geq N_0$$

where N_0 is the smallest sample size which includes observations from both sources. For example, we could define

$$\tau_N = \min \{ k \geq N_0 : |T_k| \geq A \{ N \bar{c}_k (1 - \bar{c}_k) \}^{1/2} c(\alpha) \}$$

with $c(\alpha)$ defined by (3.2), and reject H_0 if and only if $\tau_N \leq N$. This would give a test with maximum sample size N and approximate (for large N) significance level α . By taking $d_i = c_i$ in (4.13) and using Theorem 2, the properties of this test procedure under contiguous alternatives may also be studied. We omit the details. Lai (1975, page 844) has studied a sequential two-sample test based on linear rank statistics of the form

$$L_N = \sum_{i=1}^N c_i a_N(R_{Ni}).$$

At this stage it is perhaps worthwhile pointing out that a statistic such as T_N above is not well-defined in a fixed sample size setup since there is no *natural* ordering of the data which may be exploited to yield serial ranks.

5.4 The Theil statistic

Consider the simple linear regression model

$$X_i = \beta c_i + \eta_i; 1 \leq i \leq N$$

where $c_i = i + \sum_{j=1}^{i-1} j^{-1}$ for $i \geq 2$, c_1 is arbitrary and the η_i have a common Logistic distribution with density function

$$f(x) = \exp(-x)/(1 + \exp(-x))^2; -\infty < x < \infty.$$

In this case the locally most powerful test of the hypothesis $\beta = 0$ against the alternative $\beta > 0$ is based on the statistic

$$(5.1) \quad \frac{N}{\sum_{i=1}^N c_i} \frac{f'(X_i)}{f(X_i)} = \frac{N}{\sum_{i=1}^N c_i} \frac{1 - \exp(-x)}{1 + \exp(-x)} = \frac{N}{\sum_{i=1}^N c_i} (2F(X_i) - 1)$$

where $F(x) = (1 + \exp(-x))^{-1}$ is the distribution function common to the η_i . By Lemma 4.2 and (4.9) the projection of this into the class of linear serial rank statistics is given by

$$\begin{aligned} & \sum_{i=2}^N (i+1) \left(2 \frac{R_{ii}}{i+1} - 1 \right) \\ &= 2 \sum_{i=2}^N R_{ii} - \frac{(N-1)(N+4)}{2} \\ &= \sum_{1 \leq i < j \leq N} \sum_{1 \leq i < j \leq N} \text{sign}(X_j - X_i), \end{aligned}$$

after some algebra. This last quantity is the well-known Theil statistic which is used in nonparametric trend detection. Thus we see that the Theil statistic may be obtained as the projection of a locally most powerful parametric test statistic into the class of linear serial rank statistics. Somewhat more generally it may be shown that if $c_i = i + o(i)$, then the projection into the linear serial rank statistics of the statistic (5.1) is equal to

$$\sum_{1 \leq i < j \leq N} \sum_{1 \leq i < j \leq N} \text{sign}(X_j - X_i) + o(N^{3/2}), \text{ almost surely if } \beta = 0.$$

6. The independence hypothesis

Let P be a probability measure under which the random vectors $(X_{11}, X_{21}), (X_{12}, X_{22}), \dots$ are independently and continuously distributed with

$$(6.1) \quad P(X_{1i} \leq x_1, X_{2i} \leq x_2) = P(X_{1i} \leq x_1) \cdot P(X_{2i} \leq x_2) = F_1(x_1) \cdot F_2(x_2).$$

For $j > i$ we let $R_{ji} = \sum_{k=1}^j I(X_{1k} \leq X_{1i})$ and $Q_{ji} = \sum_{k=1}^j I(X_{2k} \leq X_{2i})$. We also let $U_{ri} = F_r(X_{ri})$ for $i = 1, 2, \dots$ and $r = 1, 2$ and we denote the order statistics of $U_{r1}, \dots, U_{rj}$ by $U_{rj}^{(1)}, \dots, U_{rj}^{(j)}$ for $r = 1, 2$. Finally, L_N denotes the class of statistics of the form $\sum_{i=2}^N g_i(R_{ii}, Q_{ii})$ with g_i square integrable under P .

We consider the statistics

$$(6.2) \quad T_N = \sum_{i=2}^N a_{1i}(R_{ii}) a_{2i}(Q_{ii})$$

where for $r = 1, 2$ the sequence $\{(a_{ri}(i) : 1 \leq i \leq j) : j \geq 1\}$ satisfies

$$\lim_{i \rightarrow \infty} \int_0^1 (a_{ri}(1 + [ui]) - \phi_r(u))^2 du = 0$$

$$(6.3) \quad \int_0^1 a_{ri}(1 + [ui]) du \equiv \frac{1}{i} \sum_{j=1}^i a_{ri}(j) = 0$$

with

$$(6.4) \quad A_r^2 \equiv \int_0^1 \phi_r^2(u) du < \infty \quad \text{and} \quad \int_0^1 \phi_r(u) du = 0.$$

Lemma 6.1

For $N \geq 1$ let

$$(6.5) \quad T_N^\phi = \sum_{i=1}^N \phi_1(U_{1i}) \phi_2(U_{2i})$$

and let (6.1) through (6.4) all hold. Then

$$(6.6) \quad T_N = T_N^\phi + o(N^{1/2})$$

in probability under P . Furthermore

$$(6.7) \quad (T_{(Nt)})/N^{1/2} A_1 A_2, t \geq 0 \Rightarrow_P (W(t), t \geq 0)$$

where

$$(Nt) = \min(k \geq 1 : k \geq Nt).$$

Proof

Using the method of Lemma 4.2 above it is easy to show that the projection of T_N^ϕ into L_N is given by

$$(6.8) \quad \hat{T}_N^\phi = \sum_{i=2}^N \frac{1}{i-1} a_i^{\phi_1}(R_{ii}) a_i^{\phi_2}(Q_{ii})$$

where $a_i^{\phi_r}(k) = \xi(U_{ri}^{(k)})$ for $k \leq i$, $i \geq 2$ and $r = 1, 2$. Equation (6.6) now follows by a method similar to that used in deriving (4.10). Similarly, (6.7) may be derived by following the pattern of proof established in Lemma 4.3. ■

Next, let $Q_{N,\Delta}$ for every real Δ and integer N , be a probability measure under which the random vectors $(X_{11}, X_{21}), (X_{12}, X_{22}), \dots$ are independently and continuously distributed with

$$(6.9) \quad Q_{N,\Delta}(X_{1i} \leq x_1, X_{2i} \leq x_2) \\ = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \left\{ 1 + \frac{\Delta}{N^{1/2}} g_{1N}(F_1(x)) g_{2N}(F_2(y)) \right\} dF_1(x) dF_2(y)$$

where F_1 and F_2 are defined in (6.1) and have finite Fisher information

$$(6.10) \quad I_r = \int_{-\infty}^{+\infty} \frac{f_r'(x)^2}{f_r(x)} f_r(x) dx < \infty$$

with f_r denoting the density function of F_r for $r = 1, 2$. Letting for $r = 1, 2$

$$(6.11) \quad \phi(u, f_r) = -f_r'(F_r^{-1}(u)) / f_r(F_r^{-1}(u)), 0 < u < 1$$

the functions g_{1N} and g_{2N} in (6.9) are assumed to satisfy

$$(6.12) \quad \sup \{ g_{1N}(u) g_{2N}(v) : 0 \leq u, v \leq 1 \} = o(N^{1/4})$$

$$(6.13) \quad \int_0^1 g_{rN}(u) du = 0$$

$$\lim_{N \rightarrow \infty} \int_0^1 (g_{rN}(u) - \phi(u, f_r))^2 \cdot du = 0 \quad (r = 1, 2).$$

The above setup corresponds quite closely to that used by Behnen (1971).

Now from (6.9) it is clear that the likelihood ratio for the observations $(X_{11}, X_{21}), \dots, (X_{1k}, X_{2k})$ (i.e. the restriction of $\frac{dQ_{N,\Delta}}{dP}$ to the σ -field generated by these observations) is

$$L_{N,k} = \prod_{i=1}^k \left\{ 1 + \frac{\Delta}{N^{1/2}} g_{1N}(U_{1i}) g_{2N}(U_{2i}) \right\}.$$

Following Behnen (1971), with the aid of (6.12) and a Taylor expansion we obtain for every $N \geq 1$ and $t \geq 0$,

$$(6.14) \quad \log L_{N,(Nt)} = \Delta S_{(Nt)} / N^{1/2} - \Delta^2 t I_1 I_2 / 2 + o(1)$$

in probability under P , with

$$(6.15) \quad S_k = \sum_{i=1}^k g_{1N}(U_{1i}) g_{2N}(u_{2i}).$$

From (6.13) and the Lindeberg-Feller theorem it follows that $\Delta S_{(Nt)} / N^{1/2}$ is asymptotically normally distributed with mean 0 and variance $\Delta^2 t I_1 I_2$. This together with

(6.14) shows that $\log L_{N,(Nt)}$ is asymptotically normally distributed with mean $-\Delta^2 t I_1 I_2 / 2$ and variance $\Delta^2 t I_1 I_2$. Consequently, cf. Hajek & Sidak (1967, corollary V.1.2), the sequence of $Q_{N,\Delta}$ -distributions of vectors $(X_{11}, X_{21}), \dots, (X_{1(Nt)}, X_{2(Nt)})$ is, for every $t \geq 0$, contiguous to the sequence of their P -distributions.

Theorem 3

Let (6.3), (6.4), (6.9), (6.10), (6.12) and (6.13) hold. Then, for every Δ ,

$$(T_{(Nt)} / N^{1/2} A_1 A_2, t \geq 0) \xrightarrow{Q_{N,\Delta}} (W(t) + \Delta t e_1 e_2, t \geq 0)$$

where

$$e_r = \int_0^1 \phi_r(u) \phi(u, f_r) du / A_1 A_2 \quad (r = 1, 2).$$

We omit the proof since it differs only in minor detail from that of Theorem 4. The generalization of Theorem 3 to $k (> 2)$ -dimensional data $(X_{i1}, \dots, X_{ik}), i = 1, 2, \dots$ is straightforward.

The usual rank statistic for the independence hypothesis is

$$L_N^\phi = \sum_{i=1}^N a_N^{\phi_1} (R_{Ni}) a_N^{\phi_2} (Q_{Ni})$$

where $a_N^{\phi_r}(k)$ is defined just below (6.8). By using methods similar to those in section 5 it may be shown that

$$\max \{ |L_k^\phi - \hat{T}_k^\phi| ; 2 \leq k \leq (Nt) \} = o(N^{1/2})$$

in probability under $Q_{N,\Delta}$ where $\hat{T}_N^\phi$ is given in (6.8). This, together with Theorem 3, gives the following invariance principle for the usual linear rank statistics:

$$(L_{(Nt)}^\phi / N^{1/2} A_1 A_2, t \geq 0) \Rightarrow_{Q_{N,\Delta}} (W(t) + \Delta e_1 e_2 t, t \geq 0)$$

7. Appendix

The following result extends a weak convergence theorem of Gikhman & Skorokhod (1969, section IX.3) to $D[0, \infty)$. Since the proof follows exactly the pattern indicated there, we omit the details.

Theorem 4

Let $Y_1, Y_2, \dots$ be a sequence of independent random variables with zero means and finite variances. Define for $t > 0$

$$\ell_N(t) = \min \{ k \geq 1 : \sum_{i=1}^k \text{var } Y_i \geq t \cdot \sum_{i=1}^N \text{var } Y_i \}$$

and assume that $\sum_{i=1}^N \text{var}(Y_i) \rightarrow \infty$ and that

$$\left\{ \sum_{i=1}^N \text{var } Y_i \right\}^{-1} \ell_N(t) \sum_{i=1}^{\ell_N(t)} Y_i^2 I \left\{ |Y_i| > \varepsilon \left\{ \sum_{i=1}^N \text{var } Y_i \right\}^{1/2} \right\} \rightarrow 0$$

for every $\varepsilon > 0$ and $t > 0$. Then

$$\left\{ \left\{ \sum_{i=1}^N \text{var } Y_i \right\}^{-1/2} \sum_{i=1}^{\ell_N(t)} Y_i, t \geq 0 \right\} \Rightarrow (W(t), t \geq 0).$$

REFERENCES

- BEHNEN, K. (1971). Asymptotic optimality and ARE of certain rank-order tests under contiguity. *Ann. Math. Statist.* **42** 325-329.
- GIKHMAN, I.I. and A.V. SKOROKHOD (1969). *Introduction to the theory of random processes*. W.B. Saunders.
- HÁJEK, J. and Z. SIDÁK (1967). *Theory of rank tests*. Academic Press, New York.
- LAI, T.L. (1975). On Chernoff-Savage statistics and sequential rank tests. *Ann. Statist.* **4** 825-845.
- LINDVALL, T. (1973). Weak convergence of probability measures and random functions in the function space $D[0, \infty)$. *J. Appl. Prob.* **10** 109-121.
- LOEVE, M. (1963). *Probability Theory*. D. Van Nostrand.
- MASON, D.M. (1981). On the use of a statistic based on sequential ranks to prove asymptotic properties of simple linear rank statistics. *Ann. Statist.* **9**, 424-436.
- REYNOLDS, M.R. (1975). A sequential signed-rank test for symmetry. *Ann. Statist.* **3** 382-400.
- SEN, P.K. and M.N. GHOSH (1972). On strong convergence of regression rank statistics. *Sankhyā Ser. A.* **34** 335-348.
- SEN, P.K. and M.N. GHOSH (1973). A law of the iterated logarithm for one-sample rank order statistics and an application. *Ann. Statist.* **1** 568-576.
- SEN, P.K. (1977). Tied-down Wiener process approximations for aligned rank order processes and some applications. *Ann. Statist.* **5** 1107-1123.
- SEN, P.K. (1978a). Invariance principles for linear rank statistics revisited. *Sankhyā Ser. A.* **40** 215-236.
- SEN, P.K. (1978b). Invariance principles for rank discounted partial sums and averages. *Z. Wahr. verw. Gebiete.* **42** 341-352.

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